BREAST CANCER

Peripheral blood transcriptional profiling predicts tumor subtype and neoadjuvant chemoimmunotherapy outcomes in human breast cancer

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Dynamic biomarkers of therapy response are critical for precision oncology but often rely on serial tissue biopsies, which are invasive and not always feasible. In contrast, peripheral blood offers a minimally invasive, dynamic window into the evolving systemic immune landscape. Leveraging this, we performed RNA sequencing on 546 peripheral blood samples from 160 patients with high-risk stage II/III human epidermal growth factor receptor 2 (HER2)–negative breast cancer treated with either chemotherapy alone or in combination with immunotherapy (chemoimmunotherapy). Our analysis uncovered immune correlates of tumor subtype and treatment response. For example, samples from patients with triple-negative breast cancer exhibited elevated T cell receptor clonality and robust immune activation profiles. Among patients receiving chemoimmunotherapy, early responders demonstrated high baseline T cell receptor diversity, followed by rapid clonal expansion and activation of T cells after just one treatment cycle. We developed a multiparametric peripheral immune biomarker that integrated baseline and early on-treatment features to predict response to pembrolizumab, which was successfully validated in an independent cohort of 59 patients with breast cancer treated with neoadjuvant dostarlimab. These findings reveal the potential of blood-based immune monitoring to predict immunotherapy benefit, offering an accessible tool for tailoring treatment strategies in breast cancer.

INTRODUCTION

Breast cancer (BC) remains one of the leading causes of cancer and cancer-related death among females in the United States. For localized disease, BC subtype guides the standard systemic therapy administered. Patients with hormone receptor (HR)–positive (HR^+) BC are commonly treated with endocrine therapy, with a fraction of patients benefiting from additional chemotherapy treatment. Patients bearing human epidermal growth factor receptor 2–positive ($HER2^+$) tumors are often treated with trastuzumab-based targeted therapy. Historically, treatment options for triple-negative BC (TNBC) have been limited to chemotherapy and radiation therapy because of the absence of HR and HER2.

Recently, the landscape of TNBC treatment has undergone a paradigm shift with the emergence of immunotherapy. Immunotherapy, particularly immune checkpoint inhibitors (ICIs), has become a standard of care in the management of early-stage TNBC, improving pathological complete response (pCR) rates and event-free survival (*1*–*3*). Interest in combining ICI with chemotherapy in $HR^+/HER2^-$ BC has also received substantial attention, because recent studies demonstrated increases in the pCR rate in early high-risk $HR^+/HER2^-$ BC (*2*, *4*). Despite the promising outcomes observed in several phase 3 clinical trials, the addition of immunotherapy only benefits a limited fraction of patients, exposing many to potentially lifelong and life-threatening immune-related toxicities and undue financial burden with limited therapeutic gain. Thus, there exists a pressing need to improve current biomarkers to better optimize therapy.

Tissue-based biomarkers, such as tumor-specific human leukocyte antigen (HLA)–II expression, have been validated retrospectively in a large number of studies across multiple groups (*5*–*10*). However, other biomarkers, such as program cell death ligand–1 (PD-L1) immunohistochemistry and tumor-infiltrating lymphocytes (TILs), have generally been found to be prognostic for chemotherapy benefit, limiting their use for immunotherapy-specific outcomes (*1*, *3*, *11*, *12*). Although tumor biopsy tissue remains a valuable source for biomarker identification, its utility is challenged by its invasive nature and the difficulty of obtaining longitudinal samples. Longitudinal sampling, such as before and shortly after therapy initiation, can provide deeper insight into patient-specific therapeutic response and reduce the noise of interpatient variability. Given that antitumor immunity evolves dynamically during tumor progression and treatment, the use of a liquid biopsy presents a compelling alternative. It offers a minimally invasive, repeatable source that can be leveraged to assess the evolving systemic immune response in real time and better personalize therapy.

Peripheral immunological biomarkers have already demonstrated predictive utility across several cancers. In melanoma and non–small cell lung cancer, the abundances of specific immune cell subsets and neutrophil/monocyte-to-lymphocyte ratios have been shown to correlate with response and prognosis (*13*–*16*). Yet, the development of

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more refined and predictive peripheral biomarkers demands a deeper understanding of systemic immune behavior during therapy.

To address these gaps, we performed bulk RNA sequencing (RNA-seq) on a discovery set of 546 longitudinal buffy coat blood samples obtained from 160 patients enrolled in the I-SPY2 paclitaxel control arm and "pembro4" (pembrolizumab and paclitaxel) treatment arm (followed by standard doxorubicin and cyclophosphamide in both arms). Our analysis focused on identifying peripheral transcriptomic profiles and immune cell compositions associated with BC subtype, treatment regimen, and treatment response. Additionally, we investigated the correlation between peripheral immunity and the tumor immune microenvironment using matched baseline tumor biopsies from the original report (*17*). Our findings suggest that a composite assessment of peripheral blood gene expression may hold promise as a dynamic biomarker for deployment in future immunotherapy trials and highlight the potential of transcriptional profiling as a liquid biopsy to capture systemic immune responses to cancer therapy in a minimally invasive manner.

RESULTS

Longitudinal peripheral RNA profiling enables robust immune response analyses

We processed 546 longitudinal buffy coat blood samples from 160 patients with high-risk stage II/III HER2$^-$ BC enrolled in the I-SPY2 clinical trial (table S1). Buffy coat blood samples were obtained at baseline (T0), after one cycle of therapy (T1; early treatment), between the study treatment and the standard universal treatment of doxorubicin and cyclophosphamide (T2; InterReg), and therapy end point (T3; end point/surgery) (Fig. 1A). There were 89 patients with HR$^+$ BC and 71 patients with TNBC. Each group was further subdivided into those receiving neoadjuvant paclitaxel (control; Chemo) and those receiving neoadjuvant paclitaxel and pembrolizumab (Chemo+Pembro). Treatment details and patient selection criteria have previously been reported (*18*).

Patient responses were classified as either pCR or residual disease (RD). Among the patients with HR$^+$ BC, 54 were treated with Chemo, with 8 achieving pCR and 46 having RD (14.8% pCR rate versus 13% reported previously). Thirty-five patients with HR$^+$ BC received Chemo+Pembro, with 12 achieving pCR and 23 having RD (34.2% pCR rate versus 30% reported previously in the entire treated population). For the TNBC group, 47 patients were treated with Chemo, with 10 achieving pCR and 37 having RD (21.2% pCR rate versus 22% reported previously). For patients with TNBC treated with Chemo+Pembro, there were 24 patients, with 17 achieving pCR and 7 having RD (70.8% pCR rate versus 60% reported previously) (Fig. 1B). Compared with the previously published clinical results from the I-SPY2 study (*2*), our analyzed cohort shows a slight enrichment for pembrolizumab responders.

In the biomarker validation cohort, we performed RNA-seq on 139 longitudinal buffy coat blood samples from 77 patients with HR$^+$ or TNBC receiving oral paclitaxel, encequidar, carboplatin, and dostarlimab (Dostar arm). The treatment details and patient selection criteria have been reported (*19*). There were 59 patients [20 HR$^+$ BC (7 pCRs) and 39 TNBC (23 pCRs)] with paired baseline and early-treatment samples.

Baseline peripheral immune signatures differ by BC subtypes

To determine how the immune signatures differ by BC subtype, the peripheral immune landscape was first analyzed at baseline. We performed differential gene expression analysis between baseline samples from patients with TNBC and HR$^+$ BC, followed by gene set enrichment analysis (GSEA) with Hallmark pathways. We observed an enrichment (GSEA permutation test, adjusted $P < 0.05$) of "inflammatory response" and "TNFα signaling" in patients with TNBC compared with HR$^+$ BC (Fig. 2A). Similar up-regulation of immune response signals was observed when comparing between molecular subtypes, with higher inflammatory and interferon-γ (IFN-γ)

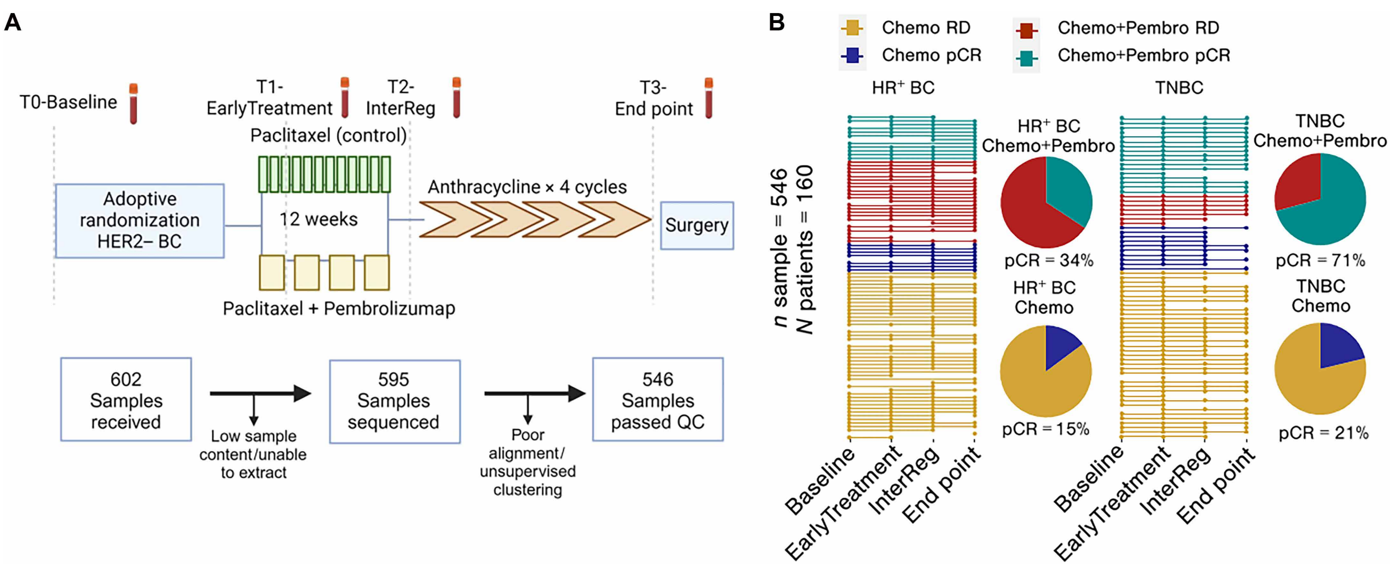

Fig. 1. I-SPY2 peripheral blood collection time points and sample distribution. (**A**) Longitudinal buffy coat blood samples from patients enrolled in the I-SPY2 Chemo control arm and Chemo+Pembro treatment arm were collected at baseline, after one cycle of therapy (EarlyTreatment), after tested regimen, and before anthracycline (InterReg) and end point. (**B**) In total, 546 RNA samples from 160 patients passed quality control (QC). Samples involved in the subsequent analysis are depicted here with patient pCR rates separated by receptor type and treatment arm. Schematic generated with BioRender.

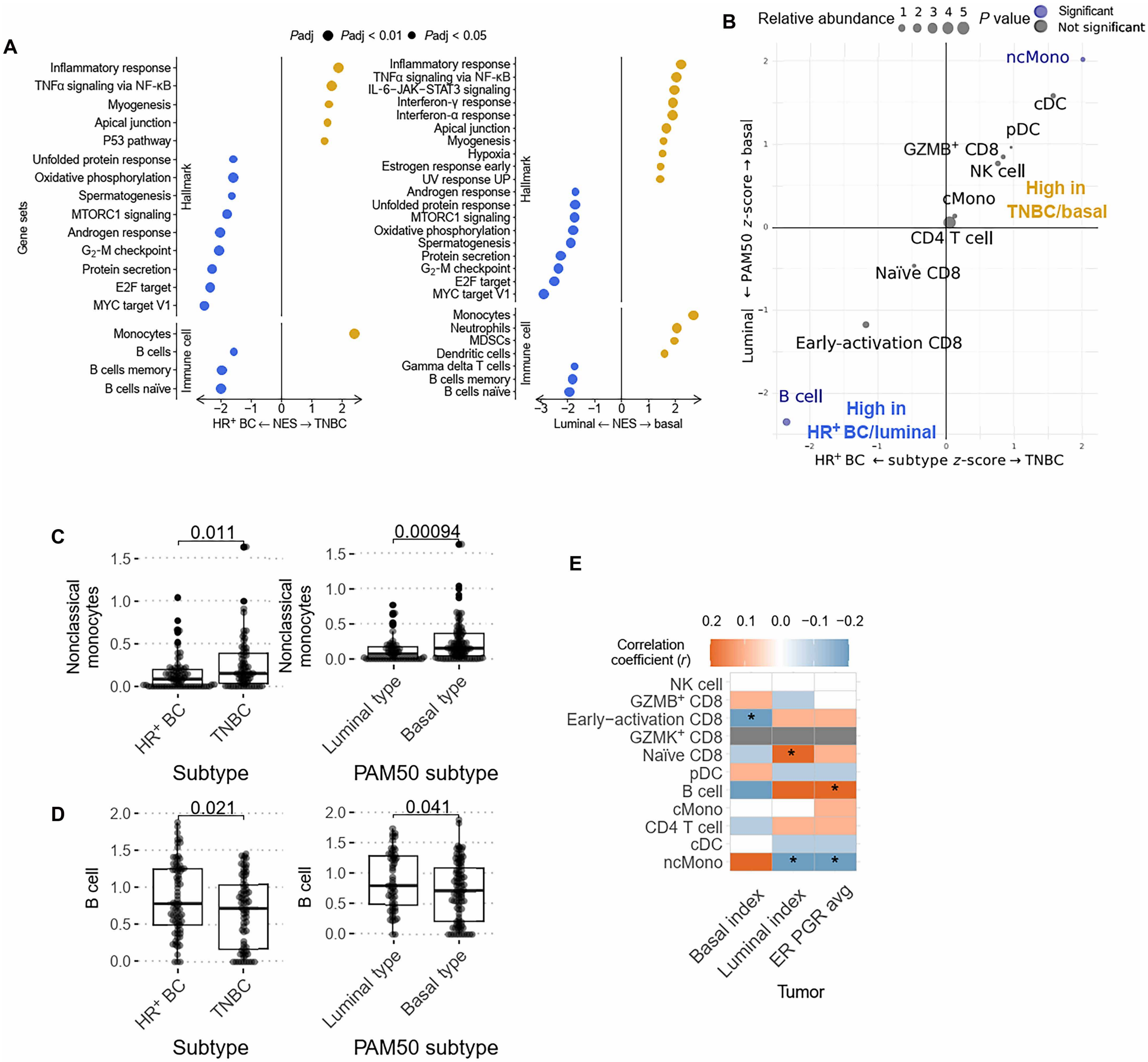

Fig. 2. Baseline peripheral immune profiles are altered among BC subtypes. (**A**) Significantly enriched hallmark gene sets and peripheral immune cell phenotype gene sets in HR$^+$ BC (blue, left; $n = 79$) versus TNBC (gold, left; $n = 69$) and in luminal-like BC (blue, right; $n = 60$) versus basal-like BC (gold, right; $n = 89$). Only pathways with GSEA permutation test adjusted P value (Padj) of <0.05 are presented here. Benjamini-Hochberg (BH) P value adjustment was performed. NES, normalized enrichment score; TNFα, tumor necrosis factor–α; NF-κB, nuclear factor κB; IL-6, interleukin-6; JAK, Janus kinase; UV, ultraviolet; MTORC1, mammalian target of rapamycin complex-1. (**B**) Association between CIBERSORT-derived immune cell abundance and Prediction Analysis of Microarray-50 (PAM50)/clinical subtype calculated with logistic regression. A positive logistic regression z-score for PAM50 (y axis) or clinical subtype (x axis) indicates that the cell type is more abundant in basal-like tumor or TNBC tumor, respectively. A negative z-score indicates that the cell type is more abundant in luminal-like or HR$^+$ BC. cDC, conventional dendritic cells; pDC, plasmacytoid dendritic cells; cMono and ncMono, classical and nonclassical monocytes, respectively. (**C** and **D**) CIBERSORT deconvoluted nonclassical monocyte (C) and B cell (D) abundances were compared between BC clinical (left: TNBC = 69 and HR$^+$ BC = 79) and PAM50 subtypes (right: basal type = 89 and luminal type = 60) by t test. Numerical P values are reported. Box plots depict median ± interquartile range (IQR) with whiskers depicting range. (**E**) The association of immune cell abundance and lineage phenotype and tumoral HR expression demonstrated by Pearson correlation. Correlations with a numerical P value of <0.05 are highlighted with an asterisk.

response signals observed in patients with basal-like BC (adjusted $P < 0.05$). Although TNBC/basal-like tumors are known to be more immunogenic (*20*, *21*), this observation was unexpected to observe in peripheral blood cells because tumor-associated immune activation is typically confined to the tumor microenvironment and is often too subtle and heterogeneous to produce a distinct, tumor subtype–specific immune signature in circulating cells. To better understand the underlying immune cell composition differences, we estimated peripheral immune cell subsets by performing GSEA using immune cell reference genes from the PanglaoDB single-cell database (https://panglaodb.se/). We observed a significant enrichment (GSEA permutation test, adjusted $P < 0.05$) of myeloid cell genes, such as those associated with monocytes, neutrophils, and myeloid-derived suppressor cells (MDSCs) in patients with TNBC/basal-like BC, whereas B cell–associated genes were up-regulated in patients with HR^+ BC/luminal BC (Fig. 2A). In addition, we applied the Cell-type Identification by Estimating Relative Subsets of RNA Transcripts (CIBERSORTx) immune cell deconvolution algorithm using our previously published BC peripheral blood mononuclear cell (PBMC) single-cell RNA-seq data (*22*) as reference to further validate the differences in PBMC composition. This analysis corroborated the findings from GSEA, with B cells significantly more abundant in the PBMCs from HR^+/luminal-like BC ($P = 0.021$), whereas nonclassical monocytes were significantly higher in TN/basal-like BC ($P = 0.011$) (Fig. 2, B to D). Given that both BC molecular subtype and clinical subtype were derived from a continuous scale of basal/luminal transcriptomic signatures or HR expression, respectively, we also calculated the correlation between PBMC immune cell abundance, basal/luminal status, and ER/PR expression level previously published by the I-SPY2 tumor analysis (*17*). Besides B cells and monocytes, we also observed a significant positive correlation between the luminal score and the abundance of naïve CD8 T cells ($P < 0.05$), which was negatively associated with the basal lineage score. The granzyme B–positive ($GZMB^+$) cytotoxic CD8 cell abundance was positively associated with the basal score (Fig. 2E). Profiling the T cell receptor (TCR) repertoire has proven to be an important method to measure autologous or treatment-induced antitumor immune response (*23*, *24*). We compared TCR clonality between BC subtypes to assess their baseline adaptive immune response. When comparing the TCR Shannon diversity between the two subtypes, we observed a lower Shannon diversity index in patients with TNBC/basal-like BC ($P = 0.047$), suggesting a slightly greater abundance of clonal T cells in their PBMCs (fig. S1, A to C). These results suggest that patients with TNBC/basal-like tumors tend to have a more inflammatory systemic immune response, associated with higher monocytes and clonal expansion of T cells.

To test the stability of the immune signature differences between BC subtypes, we performed resampling (bootstrapping) 50 times on all available baseline samples (128 HR^+ BC and 132 TNBC; combined samples from Pembro4, Dostar, and paclitaxel control arms). Differential gene expression analysis and subsequent GSEA were performed to detect peripheral immune signatures associated with BC subtypes in each iteration, allowing us to measure the stability and variability of the differential gene analysis and reduce sample-specific bias. Although the enrichment of Hallmark pathway signatures was found to be relatively noisy during the resampling process, we found highly consistent enrichment of B cell signatures in HR^+ BC and T cell signatures in TNBC (fig. S2).

Peripheral and tumor immune landscapes are correlated

Previous I-SPY2 studies sought to characterize pretreatment tumor signaling and immune landscape features using a series of mechanistically associated biomarkers (*17*). In that study, eight immune phenotype scores were generated to describe the tumor immune microenvironment, many of which were associated with treatment outcome. To further investigate the association between tumor and peripheral immune responses, we performed a correlation analysis between blood-based immune signatures and the eight continuous tumor-based immune biomarkers derived from patient-matched baseline tumor biopsy and peripheral blood samples.

Peripheral immune signatures were quantified using immune-related pathways from Hallmark pathways and immune cell signatures from PanglaoDB. Our analysis revealed that IFN response in the blood was positively associated with positive response predictors such as tumor chemokine signaling, T and B cell scores, signal transducer and activator of transcription 1 (STAT1) signaling, and tumor IFN scores that were previously defined using the tumor transcriptomic profile (Fig. 3A) (*17*). Conversely, a negative correlation was observed between the mast cell score and several peripheral immune signals, including monocyte abundance, inflammatory response, interleukin-6–STAT3 signaling, and complement signaling. This observation further supported the concordance in immunological signals between the tumor and periphery.

To classify a patient's tumor immune landscape with the eventual goal of developing a tumor-based transcriptomic signature to predict therapeutic response, the previous I-SPY2 biomarker study also developed the "Immune^+" categorical variable to identify immunogenic tumors (*17*). In brief, the Immune^+ definition was based on a variety of different subtype-specific signatures (e.g., B cell signature in HR^+ BC and T-B cell/STAT1/chemokine signature in TNBC). Patients defined as Immune^+ demonstrated higher pCR rates in the Chemo+Pembro arm (*17*). We next aimed to determine whether patients with Immune^+ tumors also exhibited a distinct peripheral immune landscape. Using differential gene expression analysis and GSEA on Hallmark pathways, adjusted for clinical subtype, we found enrichment in multiple immune response pathways, including IFN, inflammatory, and interleukin responses, in patients with Immune^+ tumors (adjusted $P < 0.05$) (Fig. 3B). Additionally, these patients had higher T cell and monocyte signatures in the peripheral blood (adjusted $P < 0.05$) (Fig. 3B). However, TCR clonality was not different between patients with Immune^+ and Immune^- tumors (fig. S1C). We then performed unsupervised clustering of PBMC composition on the basis of the score of 18 peripheral immune cell subtypes to simplify the deconvolution of the peripheral immune landscape. Seven distinct clusters (peripheral immune cell score; PICS-7) were identified on the basis of the relative enrichment score of T cells, myeloid cells, B cells, and natural killer (NK) cells (Fig. 3C; fig. S3, A to E; and table S2). When comparing the differences in PBMC immune composition subtypes (PICS-7) between $\text{Immune}^\pm$ patients, we found a numerically higher number of patients classified into B cell–enriched clusters (myeloid–B cell and NK–T–B cell cluster) in patients with HR^+/Immune^- tumors (Fig. 3D). In contrast, T cell–enriched clusters (T cell, NK–T cell, and myeloid–NK–T cell) were more abundant in patients with HR^+/Immune^+ tumors (Fig. 3D). In TNBC, the PBMC composition difference was less pronounced, with patients with Immune^- tumors predominating in the myeloid-NK clusters, whereas more Immune^+ patients were assigned to myeloid–NK–T cell clusters (Fig. 3D).

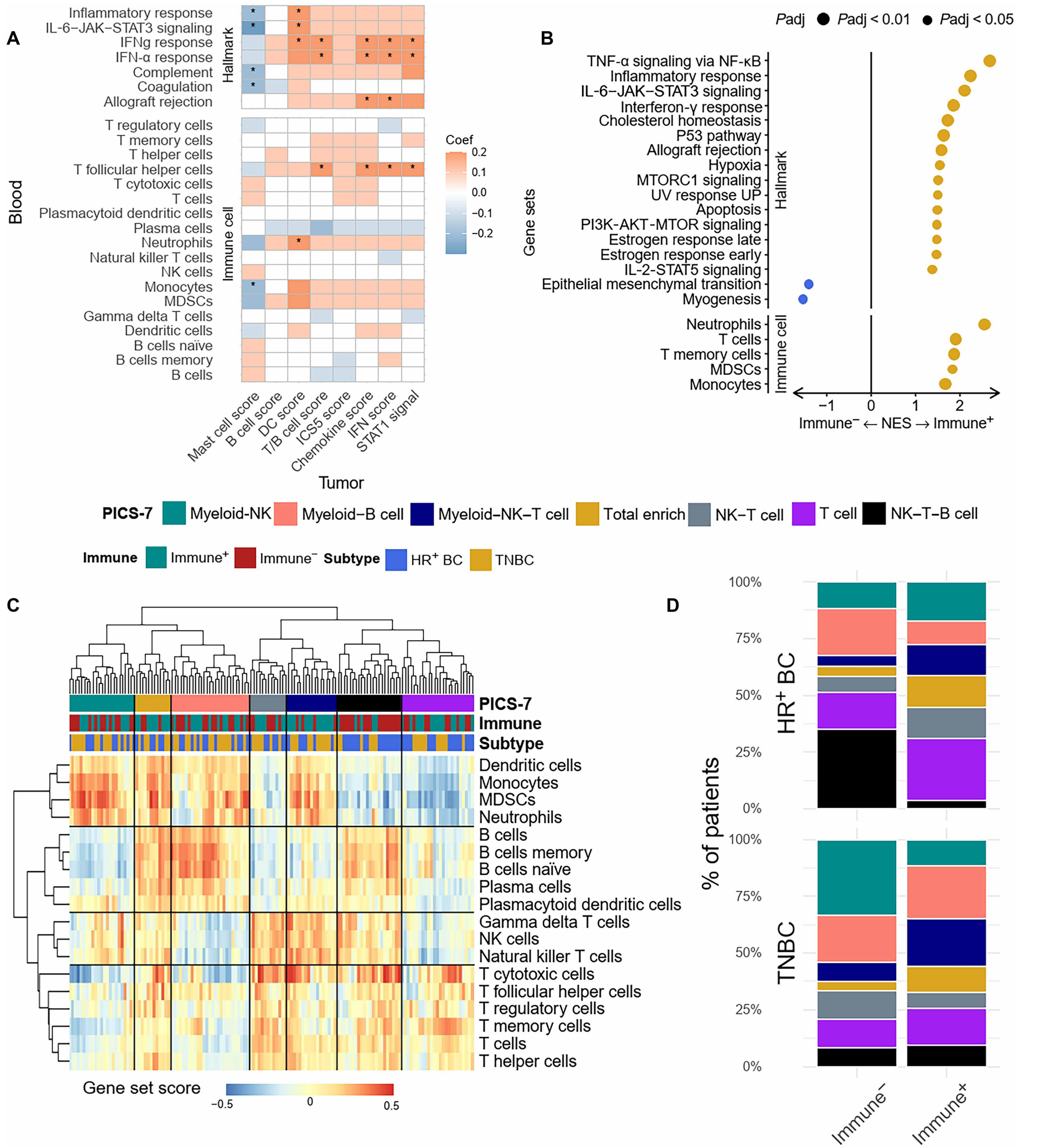

Fig. 3. Tumor immune profiles are associated with peripheral immune signatures. (**A**) Correlation analysis was performed between the peripheral blood immune signatures measured by GSVA of Hallmark pathways and immune cell subtype, and tumor immune composite scores were derived from patients' tumor microarray data. Correlation coefficients (Coef) with a numerical P value of <0.05 are highlighted with an asterisk. (**B**) GSEA was performed using the differentially expressed genes in Immune$^+$ ($n = 75$) and Immune$^-$ ($n = 73$) tumors. Significantly enriched Hallmark pathways and peripheral immune cell phenotype gene sets in Immune$^+$ (gold) and Immune$^-$ (blue) patients' blood and pathways with GSEA permutation test adjusted P value of <0.05 are presented. BH P value adjustment was performed. PI3K, phosphatidylinositol 3-kinase; MTOR, mammalian target of rapamycin. (**C**) GSVA was performed on each baseline sample to obtain the enrichment scores for different immune cell subtypes. The immune cell scores are depicted in the heatmap. Hierarchical clustering on the immune cell scores revealed seven distinct PBMC immune composition subtypes (PICS-7). (**D**) Patient baseline blood samples were classified into PICS-7. The abundance of each PICS-7 subtype in the Immune$^+$ and Immune$^-$ groups was measured separately by BC clinical subtype. Immune$^+$ TNBC = 44, Immune$^-$ TNBC = 25, Immune$^+$ HR$^+$ BC = 31, and Immune$^-$ HR$^+$ BC = 48.

We further calculated the enrichment scores of different immune cell subtype signatures in the tumors, aiming to correlate these signatures with their blood counterparts. Using ssGSEA (single-sample gene set enrichment analysis) on immune cell gene sets, we observed correlations between peripheral blood B cell abundance and tumor-infiltrating B cells, T cells, and dendritic cells [false discovery rate (FDR) of <0.1] (fig. S4). Blood T cell abundance showed modest correlations with tumor B cells, dendritic cells, and monocytes. These associations were restricted to TNBC. In HR^+ BC, none of the blood immune cell signatures correlated with their tumor counterparts (fig. S4).

Chemo+Pembro induces systemic immune activation

Longitudinal peripheral blood samples collected in this study offer an opportunity to monitor the dynamics of patients' systemic immunity during therapy. We performed a time-series analysis on the transcriptomic signatures to identify general therapeutic effects on systemic immunity induced by Chemo or Chemo+Pembro, regardless of response.

We first analyzed changes in PBMC composition using the PICS-7 clustering. After chemotherapy, 70% of patients with TNBC and 50% of patients with HR^+ BC changed PICS-7 clusters; after Chemo+Pembro, 75% of patients with TNBC and 54% of patients with HR^+ BC changed clusters (Fig. 4A). The two treatment regimens had opposite effects on B cell abundance, with Chemo+Pembro significantly reducing B cell–associated signatures in HR^+ BC as reflected by GSEA using immune cell gene sets (adjusted $P < 0.05$) (Fig. 4B). In patients with TNBC, the treatment effects on immune cell composition were largely identical between the treatment arms. We observed an up-regulation of B cell, NK cell, and T cell signatures posttreatment, accompanied by a decrease in myeloid cell abundance (adjusted $P < 0.05$) (Fig. 4, A and B). One feature of patients with TNBC treated with Chemo+Pembro was the transition of patients from the myeloid-NK phenotype to B cell– or T cell–rich phenotypes, suggesting a greater lymphocyte activation effect upon the addition of immunotherapy.

When comparing unbiased transcriptional patterns between early on-treatment versus baseline, we observed numerically more genes differentially up-regulated in Chemo+Pembro-treated patients with TNBC, followed by Chemo-treated patients with TNBC, Chemo+Pembro-treated patients with HR^+ BC, and Chemo-treated patients with HR^+ BC (Fig. 4C). There were limited overlaps among the up-regulated genes in those treatment-subtype groups, hinting at the possible existence of divergent treatment-tumor interactions in the systemic immune response. We further focused the analysis on general immune activation signatures derived from the Hallmark gene set and immune signaling gene sets previously developed by the I-SPY2 biomarker study. "STAT1 signaling" (*17*) and IFN-γ responses (Hallmark) were significantly up-regulated after Chemo+Pembro treatment ($P = 0.019$ and $P = 0.043$, respectively) (Fig. 4, D and E). In contrast, the IFN response was down-regulated after Chemo treatment ($P = 0.041$) (Fig. 4E). Both Chemo and Chemo+Pembro induced significant down-regulation of inflammatory transcriptional signals (adjusted $P < 0.05$) (fig. S5, A and B). Although these differences were significant, we also found a certain level of interpatient heterogeneity in these molecular responses to treatment, potentially contributed by the individual clinical response to therapy or patient- and disease-specific covariates.

Peripheral cytotoxic T cell signals are associated with Chemo+Pembro response in TNBC

Although we observed systemic changes induced by the two treatment regimens, patterns retained heterogeneity, potentially because of individual response to therapy. Thus, we next focused on how peripheral immune markers change during therapy in patients with pCR and RD, specifically those treated with Chemo+Pembro. By analyzing the dynamics of the peripheral immune landscape between these two groups, we aimed to identify key markers associated with therapeutic outcomes and improve our understanding of the mechanisms underlying effective immunotherapy responses.

In TNBC, NK–T cell and T cell clusters were exclusively observed in patients with pCR at baseline, with no representation in patients with RD (Fig. 5A). After one cycle of Chemo+Pembro (T1), the proportion of patients with pCR classified into the NK–T cell cluster increased numerically. In contrast, patients with RD exhibited a rise in myeloid–B cell clusters, with minimal presence of NK/T cells throughout (Fig. 5A). Further examination of detailed immune cell signatures revealed a widespread up-regulation of lymphocyte signatures after Chemo+Pembro, including genes characterizing T cells, cytotoxic T cells, and NK cells, in patients with pCR (Fig. 5B). In contrast, B cell signatures, including genes associated with naïve and memory B cells, were enriched in patients with RD posttreatment. Additionally, monocyte-, neutrophil-, and MDSC-associated genes were uniformly down-regulated in patients with pCR, whereas no significant alterations were detected in patients with RD (Fig. 5B).

One key feature observed in TNBC Chemo+Pembro responders was the enrichment of T cell signatures. Therefore, we extracted the leading-edge genes of the GSEA-enriched T cell subtypes to derive the potential immunophenotype predominating in these signatures. Among the 19 genes driving the T cell signature enrichment in patients with pCR, we found multiple markers associated with T cell effector function, such as *GZMB*, *PRF1*, *NKG7*, and *KLRD1*, and T cell maturation markers, such as *TBX21* (Fig. 5C). We next calculated the difference in these gene signatures between early treatment and baseline and found that the majority of patients with pCR experienced an up-regulation of these 19 genes (Fig. 5C). To summarize this phenotype, we calculated a composite score of these genes and observed a uniform increase in this cytotoxic T cell score in patients with pCR ($P = 0.0017$) (Fig. 5D). In contrast, the T cell composite score decreased in patients with RD after treatment ($P = 0.029$) (Fig. 5D). To determine whether this reflected a nonspecific loss of circulating T cells, we calculated the *z*-score for *CD3E/D/G*, *CD4*, and *CD8A/B*. This score was not significantly different between the two time points in patients with TNBC who did not achieve pCR after chemoimmunotherapy (fig. S6A). We then assessed the exhaustion states of T cells in nonresponders. Nonresponders experienced a consistent decrease in T cell exhaustion markers (*PDCD1*, *HAVCR2*, *TOX*, *TIGIT*, *LAG3*, and *CTLA4*) (fig. S6B) ($P < 0.05$), indicating that the decrease in the composite score was less likely to be due to exhaustion. Overall, this observation suggests that the decrease in T cell composite score in nonresponders could be due to an overall decrease in T cell abundance.

We next assessed whether enrichment of cytotoxic markers was associated with clonal T cell expansion by analyzing dynamic TCR clonality using the Shannon diversity index. We found a significantly higher level of baseline TCR diversity ($P = 0.044$) and a significant early decrease ($P = 0.031$) in diversity in patients who had a pCR on Chemo+Pembro, suggesting a more diverse T cell repertoire, followed by clonal expansion in response to treatment (Fig. 5E).

Given that T cell clonal expansion and activation was associated with response in patients with TNBC, we next calculated the T cell composite score and TCR Shannon diversity index for the two later time points: InterReg and end point (fig. S7, A and B). After the

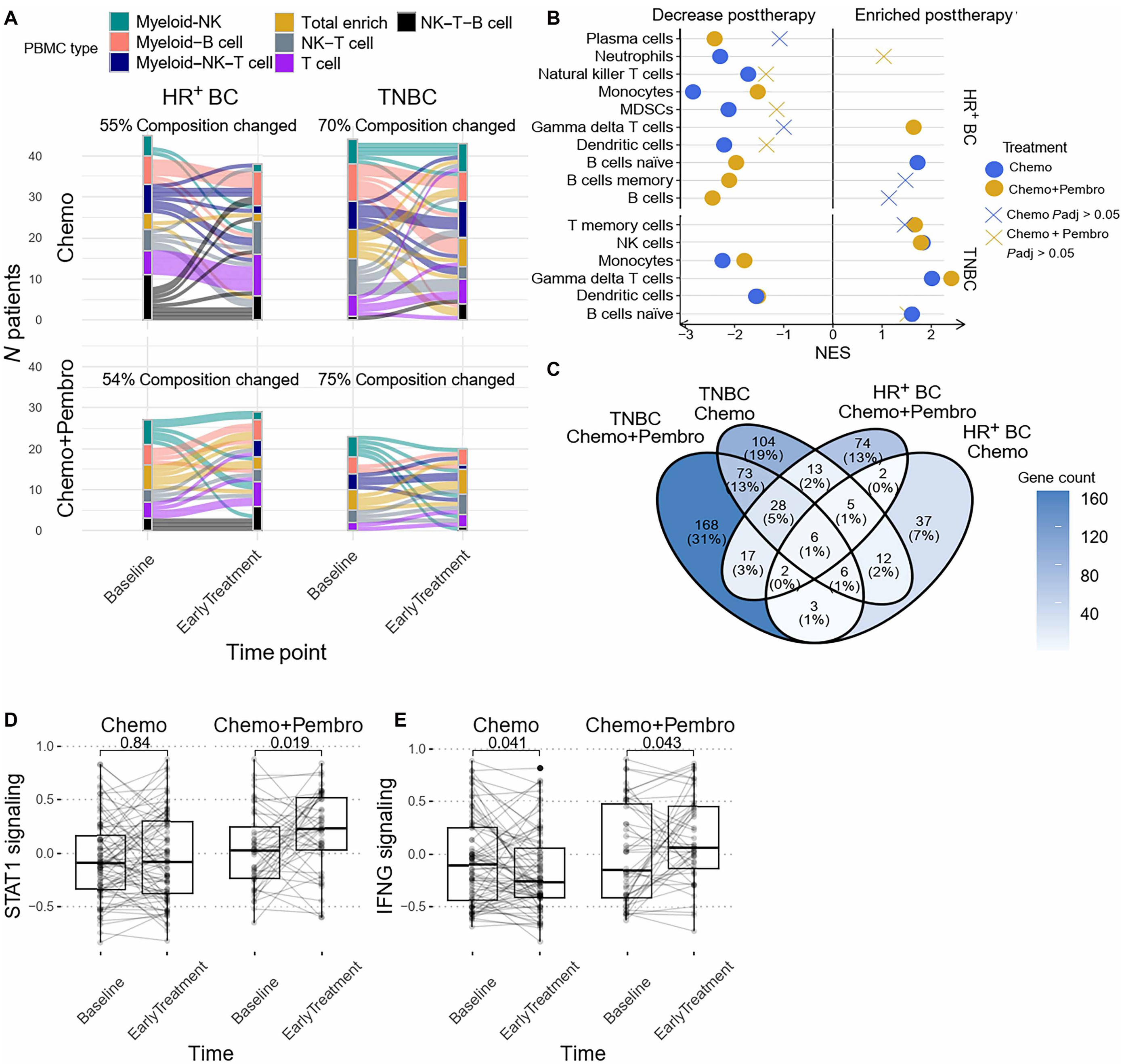

Fig. 4. Chemo and Chemo+Pembro result in divergent peripheral immune signature changes. (**A**) PBMC composition dynamics after Chemo or Chemo+Pembro treatment summarized by PICS-7 clusters, separated by BC clinical subtype. If a patient's PBMC was classified differently before and after treatment, then the patient was classified as "composition changed." The percentages of samples that experienced composition changes were measured. (**B**) Differential gene expression analysis was performed between baseline and early-treatment samples followed by GSEA using immune cell signature. A positive net enrichment score indicates that this specific immune cell expanded after treatment. The plot is separated by clinical subtype. Immune cell signatures with an adjusted *P* value of >0.05 are depicted with an X. (**C**) A Venn diagram of genes significantly up-regulated after the first round of therapy grouped by treatment and BC clinical subtype. Genes with a numerical *P* value of <0.01 and $\log_2$ fold change of >1 are considered significant. (**D** and **E**) STAT1 signaling (D) and IFN-γ signaling (E) pathways were compared before and after treatment among all patients treated with chemotherapy (left; $n = 83$) and those treated with chemotherapy and pembrolizumab (right; $n = 51$). Data were analyzed by paired *t* test. Box plots depict median ± IQR with whiskers depicting range. Samples from the same patient are connected with a line.

initial T cell clonal expansion, the T cell composite score did not change significantly.

Building on the observed T cell clonal expansion pattern post-Chemo+Pembro, we compared the early on-treatment samples between patients with pCR and RD to determine whether there were notable differences in PBMC composition. As expected, we observed significant enrichment of multiple T cell subtypes and NK cells in patients with pCR compared with patients with RD (adjusted $P < 0.05$) (fig. S8, A

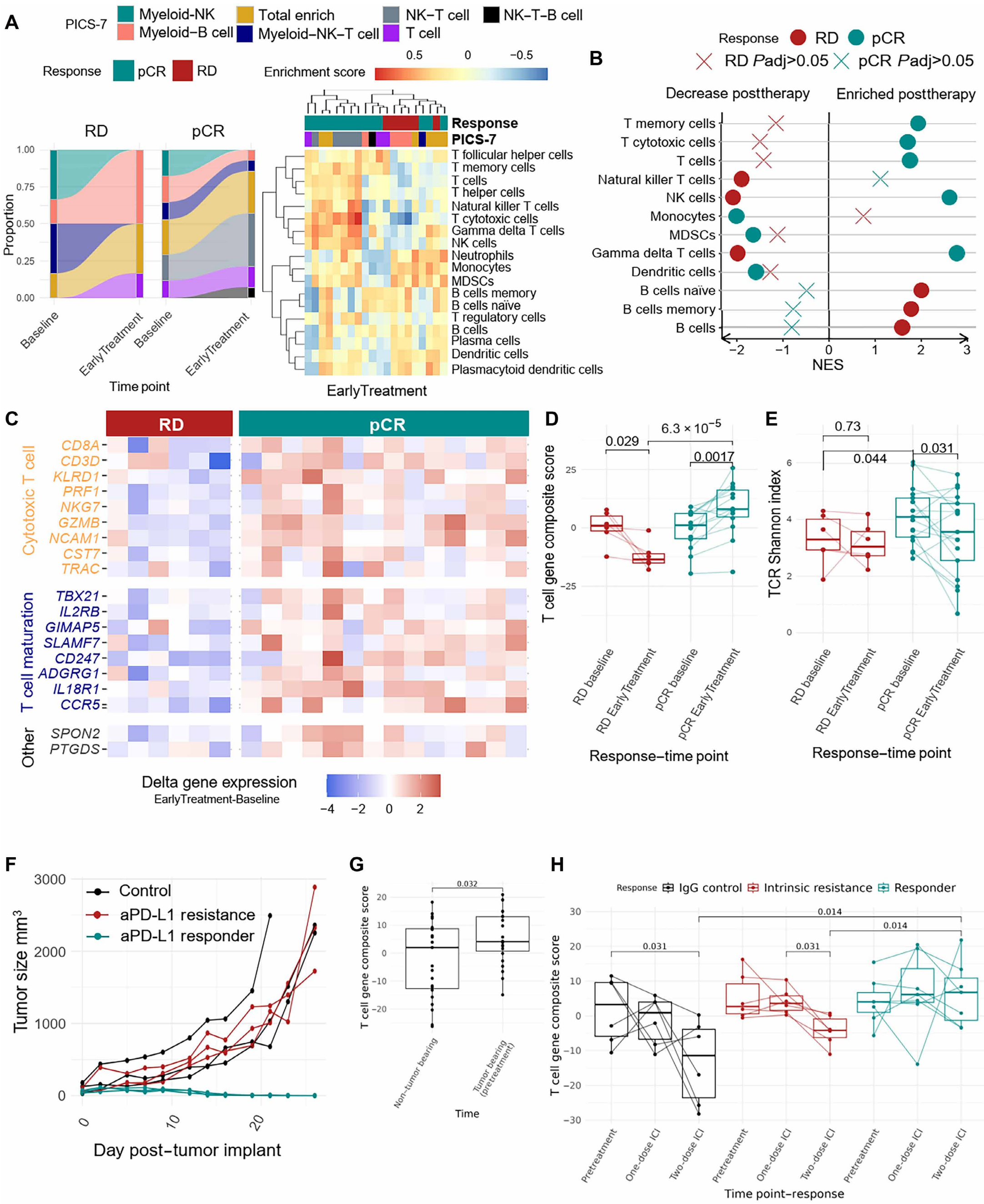

Fig. 5. Peripheral T cell activation is associated with Chemo+Pembro response in TNBC. (**A**) PICS-7 classifications for patients with TNBC were compared before and after Chemo+Pembro treatment, with the relative composition shown separately by response (left). The heatmap (right) depicts the patient gene set enrichment scores for the major immune cell subtype at early treatment. (**B**) Differential gene expression analysis was performed between baseline and early-treatment Chemo+Pembro-treated patients with TNBC followed by GSEA using immune cell signature. A positive net enrichment score indicates that the immune cell type expanded after treatment. The analysis was performed separately for patients with pCR (responders, green) and RD (nonresponders, red). (**C**) Heatmap depiction of the early treatment–baseline delta expression matrix of the leading-edge genes contributed to T cell gene sets for Chemo+Pembro-treated patients with TNBC. The leading-edge genes are grouped by function, and the plot is separated by patients with pCR (responders, green) and RD (nonresponders, red). (**D**) The 19-gene T cell composite scores in Chemo+Pembro-treated patients with TNBC ($n = 23$) were compared between baseline and early treatment, separated by response. Paired *t* tests were used to compare across time points (samples from the same patient are connected with a line), and standard *t* tests were used for comparisons between patients with pCR ($n = 17$) and RD ($n = 6$). (**E**) Baseline TCR Shannon diversity indices were compared between patients with RD ($n = 6$) and pCR ($n = 17$) and between time points as described in (D). (**F**) Example individual tumor growth curves for EMT6-bearing mice treated with anti–PD-L1 (aPD-L1) therapy. Treatments began when tumor volumes reached 100 mm^3. Curves are colored by mouse's response to therapy ($n = 3$ mice per group). (**G**) The 19-gene T cell composite score was calculated for each mouse's peripheral blood bulk RNA-seq sample. The scores were then compared between non–tumor-bearing and tumor-bearing (pretreatment) mice with paired Wilcoxon test ($n = 6$ mice per group). (**H**) The 19-gene T cell composite scores were calculated for baseline and after one dose and two doses of antibody across mice that received control IgG treatment, mice that did not respond to anti–PD-L1 therapy, and mice that responded to anti–PD-L1 therapy. Each connected dot represents a mouse ($n = 6$ mice per group). Box plots depict median ± IQR with whiskers depicting range.

and B). In contrast, imputed B cells, dendritic cells, and monocytes were significantly enriched in patients with RD (adjusted $P < 0.05$) (fig. S8, A and B). These differences in immune cell composition were sufficient to drive the clustering of patients with pCR and RD at early treatment (Fig. 5A). Of particular interest, all but one patient with pCR was assigned to the T cell–associated PICS-7 cluster, whereas the myeloid–B cell cluster was the dominant PICS-7 subtype in patients with RD. A previous study (*25*) indicated that circulating T regulatory cells and their associated proliferation markers may serve as predictors of immunotherapy response in TNBC. Here, we calculated the enrichment score of regulatory T cells between baseline and post-Chemo+Pembro and did not observe a change in regulatory T cell score in TNBC (fig. S9). The differences in quantification method (RNA-seq versus flow cytometry) may partially explain this discrepancy, given that we did not perform single-cell analysis to quantify the frequency of regulatory T cells or their phenotype.

We sought to model increases in T cell composite scores observed in patients with TNBC in mice and determine whether they were also associated with antitumor immune activity to ICIs. We used an immunotherapy responsive TNBC-like murine tumor model, EMT6. Similar to humans, tumor-bearing mice demonstrate differential responses to anti–PD-L1 treatment, ranging from complete response to intrinsic resistance, as we described in previous studies (Fig. 5F) (*26*, *27*). Longitudinal peripheral blood samples were collected at baseline (i.e., in healthy mice), after tumor implantation (i.e., tumor-bearing), and after two weekly doses of ICI treatment. Mice were followed for treatment outcome (i.e., complete response or intrinsic resistance). RNA-seq was then performed to examine peripheral immune signature changes to parallel our human studies. After tumor implantation, we observed a significant increase in the T cell composite score in the peripheral blood (adjusted $P < 0.05$), supporting our observations of a systemic immune response based on the presence of the tumor (Fig. 5G). Mice that responded to ICI treatment also demonstrated elevated levels ($P = 0.014$) of the peripheral T cell score compared with untreated mice or those demonstrating ICI resistance (Fig. 5H). Thus, these data support our clinical findings in patients and the hypothesis that systemic immune features show a direct relationship to the immune response in the tumor.

Down-regulation of leukocyte chemotaxis distinguishes Chemo+Pembro responders in HR$^+$ BC

We previously demonstrated differences between BC subtypes both at baseline and in response to therapy, suggesting that the underlying biology of the disease and its response to therapy (even at the systemic level) may result in divergent responses. In patients with HR$^+$ BC, we noted a subtle stratification of responders and nonresponders by immune cell enrichment at baseline. At baseline, the NK–T cell cluster (PICS-7 subtype) was exclusively detected in patients with pCR (Fig. 6A). Differential enrichment analysis at baseline further revealed that T cell– and NK cell–associated gene signatures were predominantly enriched in patients with pCR, whereas B cell signatures were more prevalent in patients with RD (fig. S8B).

Detailed analysis of differentially enriched immune cell signatures post-Chemo+Pembro revealed a significant decrease in myeloid cell marker expression in patients with pCR and an increase in these markers in patients with RD (adjusted $P < 0.05$) (Fig. 6B). Specifically, markers associated with monocytes, MDSCs, neutrophils, and dendritic cells (DCs) contributed to the change. Immune cell deconvolution corroborated these findings, showing a significant decrease in monocyte abundance (combining nonclassical and classical monocytes) in patients with pCR ($P = 0.03$) (Fig. 6C), whereas no significant trend was detected in patients with RD. We further explored other peripheral immunological features associated with the decrease in monocytes and found a concurrent decrease in the leukocyte chemotaxis score posttreatment in patients with pCR (Fig. 6D). Given that the leukocyte chemotaxis score was significantly correlated ($P = 0.026$) with the tumoral chemokine score at baseline (fig. S10), this observation could suggest that Chemo+Pembro may modulate the tumor microenvironment in patients with HR$^+$ BC by reducing monocyte recruitment via the peripheral compartment.

Although T cell immune signatures were not differentially enriched during therapy, we investigated whether the T cell composite score, previously defined for Chemo+Pembro-treated patients with TNBC, could reduce bias and help detect T cell changes in HR$^+$ BC during Chemo+Pembro treatment. Besides a higher baseline T cell composite score ($P = 0.0064$) (Fig. 6E) and higher baseline and early-treatment TCR Shannon diversity index ($P = 0.018$) (Fig. 6F) in patients with pCR, we did not observe significant changes in T cell–associated metrics induced by Chemo+Pembro throughout the treatment (fig. S11, A and B). Our findings indicate that distinct immune activation mechanisms may be induced by Chemo+Pembro in HR$^+$ BC compared with TNBC, highlighting the complexity and specificity of the immune response in different BC subtypes.

T cell activation status is associated with Chemo+Pembro response

Although various immune response signals and dynamics in immune cell composition were observed, only the higher baseline 19-gene T cell composite score (HR$^+$ BC specific), posttreatment enrichment of T cell composite score (TNBC specific), posttherapy reduction in leukocyte chemotaxis (HR$^+$ BC specific), and baseline TCR clonality were exclusively associated with the Chemo+Pembro response (Figs. 5, D and E, and 6, E and F; and fig. S12, A to D). To explore whether the addition of immunotherapy might benefit chemotherapy nonresponders or potentially disadvantage chemotherapy responders, we performed differential gene expression analyses on chemotherapy-treated patients, comparing responders and nonresponders at baseline and early on-treatment. Our goal was to identify changes in immune features that were either concordant or opposite to those associated with response to Chemo+Pembro.

In TNBC, chemotherapy nonresponders exhibited a reduction in monocyte abundance after treatment, accompanied by down-regulation of inflammatory and IFN-response gene programs (fig. S13A). Despite these decreases, cytotoxic T cell signatures became significantly enriched after therapy in the chemotherapy nonresponder group (adjusted $P < 0.05$). Many of these changes mirrored the immune features observed in patients who responded to chemoimmunotherapy (Fig. 5B). This pattern could provide some evidence that pembrolizumab may help "rescue" or restore antitumor immune activity in patients who otherwise do not mount a favorable immune response to chemotherapy alone, possibly by accelerating circulation and infiltration of inflammatory (IFN-γ–induced) myeloid cells.

In HR$^+$ patients, chemotherapy responders and nonresponders exhibited largely similar shifts in peripheral immune cell composition (fig. S13B). The only notable exception was B cell abundance: Nonresponders showed an increase in B cell signatures after treatment, whereas responders did not. However, among patients receiving

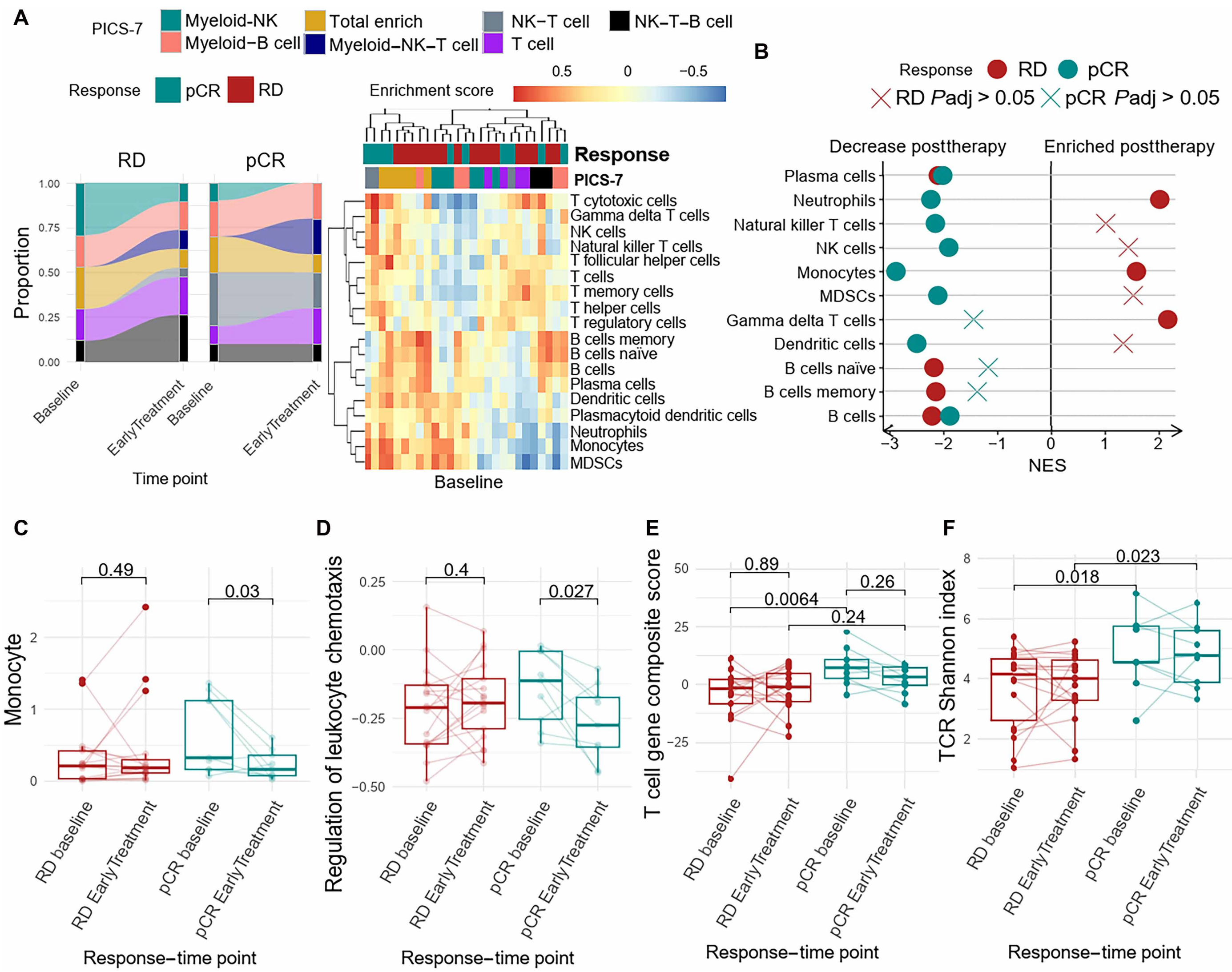

Fig. 6. Baseline T cell signatures are associated with Chemo+Pembro response in HR$^+$ BC. (**A**) HR$^+$ patients' PICS-7 classifications were compared before and after Chemo+Pembro treatment; the relative composition is measured separately by response (left). The heatmap depicts the patients' gene set enrichment scores for the major immune cell subtypes at baseline. (**B**) Differential gene expression analysis was performed between baseline and early-treatment Chemo+Pembro-treated patients with HR$^+$ BC followed by GSEA using immune cell signature. A positive net enrichment score indicates that this specific immune cell expanded after treatment. The analysis was performed separately for patients with pCR (responders, green; $n = 9$) and RD (nonresponders, red; $n = 18$). (**C**) CIBERSORT deconvoluted monocyte (classical and nonclassical) abundances were compared before and after Chemo+Pembro treatment in responders ($n = 9$) and nonresponders ($n = 18$). (**D**) Gene Ontology biological process leukocyte chemotaxis scores were calculated for each sample using GSVA. The enrichment scores were compared before and after Chemo+Pembro treatment with paired *t* test. (**E**) The 19-gene T cell composite scores in Chemo+Pembro-treated patients with HR$^+$ BC were compared between baseline and early treatment, separated by response. Paired *t* test was used to compare across time points; regular *t* test was used for comparisons between patients with pCR and RD. (**F**) Baseline TCR Shannon diversity indices were compared between patients with RD and pCR with *t* test. The TCR Shannon diversity index was also compared between baseline and early-treatment samples using paired *t* test. Box plots depict median ± IQR with whiskers depicting range.

Chemo+Pembro, all patients, regardless of response, demonstrated a decrease in B cell signatures. Although this pattern raises the possibility that pembrolizumab may "reverse" the B cell enrichment observed in chemotherapy nonresponders, these B cell changes were not associated with treatment response in HR$^+$ disease. As a result, it remains unclear whether the addition of pembrolizumab meaningfully "rescues," offsets, or, otherwise, modifies chemotherapy response in HR$^+$ BC.

To further test the predictive power of these markers, we generated a logistic regression model incorporating the T cell composite score at baseline (HR$^+$ BC), the reduction in leukocyte chemotaxis score (HR$^+$ BC), the increase in T cell composite score at early treatment (exclusive for TNBC), and the baseline TCR clonality to distinguish chemotherapy and pembrolizumab responders from nonresponders. We performed training and testing on the same dataset and found that the regression model trained on these peripheral immune matrices could effectively identify patients who responded to the treatment regimen (Fig. 7A and table S3). In HR$^+$ BC, the strongest prediction variable was the baseline T cell composite score, seconded by changes in chemotaxis score. In TNBC, the strongest predictor was the changes

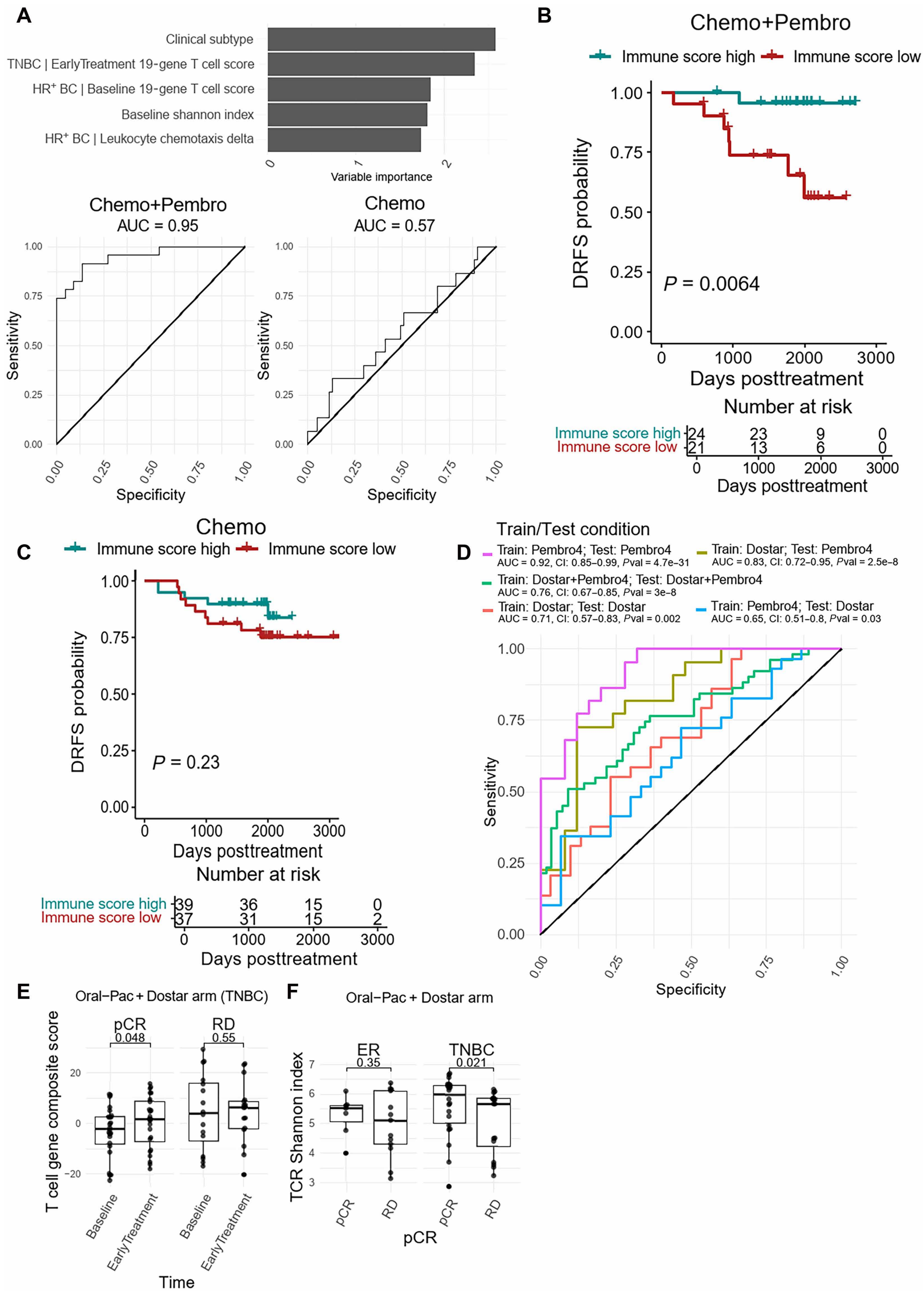

Fig. 7. Peripheral immune biomarkers predict chemoimmunotherapy response. (**A**) A logistic regression using BC clinical subtype, 19-gene T cell signature, changes in leukocyte chemotaxis score, and baseline TCR Shannon diversity index was performed to predict patients' responses to Chemo+Pembro. We used the same training and testing dataset, containing all patients treated with Chemo+Pembro. (**B** and **C**) Kaplan-Meier analysis of DRFS for the Chemo+Pembro (B) and Chemo (C) arms of the I-SPY2 trial according to the GLM prediction score based on peripheral immune signatures. Patients were classified as Immune high if the GLM obtained a probability score of ≥0.45 for Chemo+Pembro and ≥0.19 for Chemo. (**D**) ICI response prediction model validation was performed using various training and testing data combinations. The AUC for each condition is listed, along with 95% CIs and the *P* value. (**E**) The 19-gene T cell composite scores were compared for patients with TNBC in the Dostar trial before and after treatment. Paired *t* tests were performed separately for patients with pCR ($n = 24$) and RD ($n = 16$). (**F**) The baseline TCR Shannon diversity indices were compared between patients with pCR (HR^+ BC = 7 and TNBC = 24) and RD (HR^+ BC = 14 and TNBC = 16). *t* tests were performed separately for patients with HR^+ BC and patients with TNBC. Box plots depict median ± IQR with whiskers depicting range.

in T cell composite score (Fig. 7A and fig. S14, A and B). However, the model did not distinguish chemotherapy responses. Model performance was further tested with 10-fold cross-validation. The Kappa value was higher in Chemo+Pembro than Chemo (0.64 versus 0), strengthening the association between those peripheral immune biomarkers and Chemo+Pembro responses. We next tested whether the peripheral immune signatures could identify long-term benefits. We separated the patients as immune signature high versus low with an optimal cutoff from logistic regression ROC (receiver operating characteristic) curve. When comparing the patient's distance recurrence-free survival (DRFS), patients with high immune scores had significantly improved DRFS when compared with the low immune score cohort ($P = 0.0064$) (Fig. 7B). This relationship was not observed in the Chemo arm ($P = 0.23$) (Fig. 7C).

It is acknowledged that training and testing on the same dataset likely results in overfitting. We therefore performed validation on the baseline and early-treatment samples from 59 patients receiving oral paclitaxel, encequidar, and dostarlimab. The model, when using data from Pembro4 (the original discovery set) as training set, maintained predictive capacity to dostarlimab combination therapy [area under the curve (AUC) of 0.65; confidence interval (CI): 0.51 to 0.8; $P = 0.03$] (Fig. 7D). Thus, these findings demonstrate potential generalizability across multiple treatment arms.

When using data from the Dostar arm to train the coefficients of the model parameters and testing on the Pembro4 arm, the model achieved an AUC of 0.83 (CI: 0.72 to 0.95; $P = 2.5 \times 10^{-8}$). A combined training on both cohorts and testing on the pooled set achieved an intermediate AUC of 0.76 (CI: 0.67 to 0.85; $P = 3 \times 10^{-8}$), suggesting robustness of the model when leveraging a more diverse training population. Overall, these results suggest that the Pembro4-trained model captures a biologically relevant signal that is, to some extent, transferable to patients receiving a distinct but mechanistically related immunotherapy regimen. Given the observed differences in predictive capacity when calculating coefficients on different training/test populations, we anticipate that inclusion of additional data across more diverse populations of ICI-treated patients will help stabilize the model for the broadest and most reliable generalizability.

In addition, we observed an enrichment in T cell composite score in patients with TNBC responding to oral paclitaxel, encequidar, and dostarlimab, similar to the patients who responded to pembrolizumab (Fig. 7E). We also observed higher baseline TCR Shannon diversity indices in TNBC responders ($P = 0.021$) (Fig. 7F). These observations indicate that baseline as well as on-treatment T cell responses are associated with immunotherapy outcomes, and these immunological signatures could be generalized across different ICI regimens.

We also generated a TNBC-only model and a HR^+ BC–only model, each containing subtype-specific ICI response biomarkers. We found that the model performance was similar between TNBC and the original model (fig. S15A). In addition, we tested whether the model performance would be different if only incorporating baseline features such as baseline T cell composite score or TCR clonality. We found that the ROC curve for the simplified baseline model was not significantly different from the original model (fig. S15B). This result suggests that, upon further validation, there is potential to further simplify the prediction model, which will require larger cohorts of chemoimmunotherapy-treated patients.

DISCUSSION

The cancer immunity cycle (*28*) proposes that tumor-derived antigens, released during natural cell turnover or in response to therapy, drain into lymph nodes, where they are captured by professional antigen-presenting cells and presented to T cells for priming. Once appropriately activated, T cells are "licensed" to exit the lymph node and migrate toward sites of inflammation, traversing the systemic circulation as a necessary conduit. This framework provides biological rationale for sampling the peripheral immune system as a surrogate for monitoring ongoing antitumor immunity.

In this study, we investigated whether peripheral blood transcriptomic signatures, captured before and after chemoimmunotherapy (Chemo+Pembro), could reveal evidence of systemic antitumor immune activity in patients with BC. Leveraging more than 500 RNA-sequenced peripheral blood samples collected longitudinally from 160 patients, we found that the baseline systemic immune environment varied significantly across BC phenotypes, including clinical subtypes, molecular subtypes, and the tumor immune microenvironment. These baseline differences persisted and evolved during Chemo+Pembro treatment, giving rise to dynamic, subtype-specific transcriptomic patterns that were associated with therapeutic response (fig. S16).

Among subtypes, TNBC, particularly basal-like tumors, emerged as the most immunogenic. This is likely attributable to their high tumor mutation burden (*29*), elevated expression of HLA molecules (*27*, *30*), and a well-documented enrichment in TILs (*20*, *31*). These features contribute to a preexisting inflamed peripheral immune milieu that, upon effective Chemo+Pembro therapy, is further characterized by systemic T cell clonal expansion and activation, mirroring the immune dynamics observed in other immune-responsive cancers (*15*, *32*, *33*).

The recent success of the KEYNOTE-756 trial has challenged the long-standing belief that immune-cold HR^+ BC is inherently unresponsive to immunotherapy (*4*). In our study, we observed that Chemo+Pembro treatment induced minimal changes in peripheral T cell activity in patients who are HR^+, aligning with prior observations in metastatic tumors (*34*). However, Chemo+Pembro responders in this group exhibited a decrease in circulating monocyte abundance and leukocyte chemotaxis scores. This suppression of myeloid-associated signals may point to a reduction in immune suppression by limiting tumor-associated macrophages or MDSCs. Such an effect may result from diminished chemokine and cytokine signaling involved in myeloid cell recruitment to the tumor microenvironment (*35*–*37*), distinguishing it from the classical T cell–mediated immune activation seen in TNBC and other immune-hot tumors.

Our findings further demonstrate that peripheral immunophenotypes, particularly longitudinal transcriptomic changes, can serve as predictors of immunotherapy response. A well-known limitation in biomarker discovery using baseline samples is the substantial interpatient variability driven by factors such as age, sex, and disease history. In contrast, longitudinal sampling, as applied in our study through paired baseline and early-treatment blood samples, enables the identification of therapy-induced immune changes while minimizing confounding from static patient characteristics. This approach is especially relevant in immunotherapy, where both tumor and immune compartments evolve during treatment. Moving forward, future studies should prioritize multi–time point sampling strategies to uncover dynamic immune signatures associated with therapeutic response.

It has been established that the presence of malignancy can induce systemic suppression of the immune system through interactions in the bone marrow affecting myelopoiesis (*38*). Triple-negative mammary cancer models in mice have been shown to affect B cell maturation in the bone marrow, reducing the number of differentiated B cells relative to pre-pro B cells and overall B cells. Other models, like PyMT (polyoma middle-T; generally regarded as a more luminal B-like model), caused a generally opposite effect, increasing early B cells (*39*). Although analogies between murine models and human subtypes of BC exist, they are generally regarded as imperfect. Nonetheless, these prior studies suggest that tumor phenotypes guide systemic immunity, acting via the bone marrow to affect myeloid phenotypes and exhaustion states of T cells. Similar effects were found in the peripheral blood in both preclinical models and humans. Although these studies tested primarily the state of peripheral B cells before systemic therapy rather than dynamic responses to therapy, our results are aligned and thus may provide mechanistic underpinnings. Despite using different methods of assessment between our studies (flow cytometry versus transcriptional imputation), patients with a predominant myeloid–B PICS-7 cell phenotype after initiating systemic chemoimmunotherapy demonstrated high rates of RD (50% versus 10%).

Although the longitudinal peripheral blood dataset presented here offers an opportunity to study systemic immune responses under immunotherapy, our study has several important limitations. Subdividing the cohort by clinical subtype, treatment arm, and response status results in smaller subgroup sizes and uneven distributions, which, in turn, limits statistical power and the ability to control for sample-specific variation. Although we validated our ICI response prediction biomarkers in an independent I-SPY2 cohort treated with oral paclitaxel and dostarlimab, model performance varied, likely because of differences in treatment regimens, patient populations, sample handling, or batch effects. The dostarlimab arm was a limited neoadjuvant arm (combining with oral paclitaxel), but, overall, that arm did not "graduate" to a phase 3 design, suggesting that it had less activity than pembrolizumab. Therefore, this treatment regimen may have enrolled fewer patients on the regimen who responded to chemoimmunotherapy over chemotherapy alone. These caveats underscore the need for future studies that integrate larger, harmonized datasets and apply advanced machine learning or deep learning methods capable of modeling complex immune features to better stratify patients and improve predictive accuracy. Given these challenges, our statistical analyses were designed to be robust and descriptive, capturing key biological signals across heterogeneous subgroups. Although additional validation in larger, treatment-specific cohorts will strengthen the generalizability and predictive utility of individual biomarkers, the reproducibility of several key findings across independent cohorts supports the translational potential of this approach. For HR^+ BC cases, there was a lack of association between baseline tumor and peripheral blood immune cell abundances. Such discrepancy may be attributed to differences in tumor immunogenicity, given that it may decouple systemic immune signals from local tumor immune architecture, resulting in weaker or absent correspondence between blood-based and tumor-resident immune signatures. In addition, the more abundant "general highway" feature of peripheral blood may mask many more granular correlations (particularly in the absence of refined single-cell RNA-seq analysis) imputed from bulk sequencing, akin to a "needle in a haystack" hypothesis. Thus, it may be a computational and biological challenge to observe such correlations between the two environments, even if they exist. A further limitation of this study is the lack of high-resolution immune cell profiling. Although we used CIBERSORT for immune cell deconvolution, our use of buffy coat samples precluded live cell assays such as flow cytometry or single-cell RNA-seq. This limitation heavily influenced our decision to rely on bulk RNA-seq as our primary immune profiling strategy. To address this gap, we are planning future prospective studies using cryopreserved blood samples, which will enable multiomic analyses incorporating transcriptomic, proteomic, and cytometric approaches to achieve a more granular view of peripheral immune dynamics.

Despite these limitations, the dataset presented here provides a view into the interplay between systemic and tumor-localized immunity under therapeutic perturbation. The biomarkers identified, upon further validation, have the potential to serve as dynamic, minimally invasive "liquid biopsies" that can guide immunotherapy decision-making, tailor treatment regimens, and advance precision oncology not only in BC but potentially in other solid tumors as well.

MATERIALS AND METHODS

Study design

This was a retrospective exploratory analysis of all available peripheral blood buffy coat samples longitudinally collected from the control (chemotherapy alone) and pembrolizumab and chemotherapy arms of the I-SPY2 adaptive clinical trial (NCT01042379). Patient selection criteria, dosage, and response have been previously reported in detail (*19*). From serial blood samples, we generated multidimensional profiling data (RNA-seq). Multidimensional profiling data were analyzed to identify peripheral immune correlates of treatment response and resistance mechanisms of immune checkpoint blockade. The investigators of this secondary analysis were blinded to the patient characteristics and treatment arm for samples until final quality control checks on the resulting RNA-seq data.

Five hundred forty-six samples from 160 patients passed final quality control and were included in this study. Patients were treated with weekly paclitaxel for 12 weeks ± four rounds of pembrolizumab administered every 3 weeks. After initial treatment, additional neoadjuvant treatment of four cycles of doxorubicin and cyclophosphamide was administered. Patient selection criteria, dosage, and response have been previously reported in detail (*2*, *18*). Longitudinal samples were collected at baseline, after one treatment cycle (3 weeks of paclitaxel ± pembrolizumab), after initial treatment completion, and after additional neoadjuvant doxorubicin and cyclophosphamide before surgery. For the validation cohort, 139 samples from 77 patients passed final quality control. Among these samples, 59 patients had both baseline and early-treatment samples and were selected for subsequent validation study. Patients were treated with oral paclitaxel (205 mg/m^2) and encequidar (12.9 mg) on days 1 to 3, carboplatin AUC of 1.5 (targeting AUC of 1.5 mg/ml per min) on day 1 weekly for 12 weeks, and dostarlimab of 500 mg every 3 weeks for 12 weeks, followed by doxorubicin/cyclophosphamide every 2 to 3 weeks for 8 to 12 weeks. Patient selection criteria, dosage, and response have been previously reported in detail (*19*). I-SPY2 was conducted in accordance with the guidelines for Good Clinical Practice and the Declaration of Helsinki, with approval for the study protocol and associated amendments obtained from independent ethics committees at each site. Written informed consent was obtained from each participant before screening and again before treatment. The I-SPY2 Data Safety Monitoring Board meets monthly to review patient safety.

Buffy coat peripheral blood collection
Within 2 hours of blood collection, tubes containing whole blood were centrifuged at 1100*g* to 1300*g* for 20 min at room temperature. The buffy coat layer (above the red blood cells) was removed and aliquoted into four 2-ml cryovials, with some red blood cells allowed in the buffy coat. Two buffy coat vials were frozen directly upright at −80°C.

Bulk RNA-seq and data preprocessing
Total RNA was isolated from frozen buffy coat blood samples using the Maxwell 16 LEV simplyRNA Blood Purification Kits (catalog no. AS1310) following the manufacturer's instructions. Briefly, frozen buffy coat samples were thawed and went through red blood cell lysis. The remaining PBMCs were lysed and homogenized to release RNA. Magnetic beads specific for capturing RNA were used to isolate RNA from cellular debris and other contaminants. After washes to remove impurities, the purified RNA was eluted in water, and quality control was performed with NanoDrop. Next, mRNA enrichment and cDNA library were prepared using the stranded mRNA (polyA-selected) library preparation kit (Illumina). Sequencing was performed at paired-end 150 base pairs (bp) on the Illumina NovaSeq 6000 targeting an average of 50 million reads per sample. Demultiplexed FASTQ files were next aligned using STAR with a genome index generated from human Hg38. FeatureCount was next applied to create gene count matrix. Subsequent MultiQC was performed to ensure sample homogeneity. In parallel, TCR sequences were extracted from raw FASTQ files with MixCR using default parameters based on the developers' instructions (https://github.com/milaboratory/mixcr/). In brief, sequencing reads were aligned to reference V, D, J, and C genes of TCRs. Then, the aligned reads were assembled to extract CDR3 gene regions. Last, the clonotypes were exported in a human-readable/parsable tab-delimited text file format. Combat-Seq (R, sva package) was performed to remove potential batch effect from the discovery and validation dataset. The patient's BC subtype was used as a covariate during correction.

Orthotopic animal experiments
EMT6 murine (BALB/c) mammary carcinoma cells (American Type Culture Collection, CRL-2755) were cultured in Dulbecco's modified Eagle's medium/F12 (Thermo Fisher Scientific) supplemented with 10% fetal bovine serum (Thermo Fisher Scientific). For murine models, EMT6 cell lines (5×10^4 cells) were injected in 100 μl of phosphate-buffered saline into the fourth mammary fat pad of 6-week-old BALB/c female mice. Tumor formation and growth were evaluated for up to 100 days. Tumors were measured two times weekly with calipers, and volume was calculated in cubic millimeters using the formula ($\text{width}^2 \times \text{length}/2$). Mice were humanely euthanized when their tumor burden reached 2000 mm^3, at 100 days for survival studies, or at earlier time points for analysis. Blood was collected from mice via the submandibular vein before and after tumor inoculation and after every round of treatment. All animal studies were reviewed by the Vanderbilt University Medical Center Institutional Animal Care and Use Committee within the Animal Care and Use Program to monitor research integrity and compliance (protocol M2300055).

In vivo therapy regimens
Anti–PD-L1 (clone 6E11; Genentech) was administered at an initial dose of 200 μg intraperitoneally when individual mouse tumors reached 100 mm^3 and subsequently dropped to 100 μg weekly for up to 4 weeks of total administration time. The definition of response to immunotherapy has been documented in our previous study (*26*). Briefly, intrinsic resistance mouse tumor growth showed no difference compared to the immunoglobulin G (IgG) control. In contrast, complete responders showed significant tumor growth difference and complete tumor clearance at therapy end point.

Mouse peripheral blood bulk RNA-seq
RNA was harvested from mouse PBMCs using the Maxwell 16 automated workstation (Promega) using the LEV simplyRNA Tissue or Blood purification kit (Promega), respectively. A quality control analysis was performed on the RNA samples using the Agilent Bioanalyzer to assess quality, and an RNA Qubit assay was used to determine quantity. Library preparation was completed using the NEBNext rRNA Depletion kit and protocol (New England Biolabs) per the manufacturer's recommendations. The kit uses a ribonuclease H (RNAse H)–based method to deplete both cytoplasmic and mitochondrial ribosomal RNA. After ribosomal depletion, the RNA samples underwent messenger RNA enrichment using oligo(dT) beads, followed by thermal fragmentation using divalent cations under elevated temperature. First-strand cDNA synthesis was performed using random hexamer primers, whereas second-strand cDNA synthesis was carried out using DNA polymerase I and RNAse H. End repair, A-tailing, and adapter ligation were performed to generate the final cDNA library. The library quality was assessed using an Agilent Bioanalyzer and quantified using a quantitative PCR-based method with the KAPA Library Quantification Kit (Roche) and the QuantStudio 12K instrument.

Prepared libraries were pooled in equimolar ratios, and the resulting pool was subjected to cluster generation using the NovaSeq 6000 System, following the manufacturer's protocols. Paired-end sequencing (150 bp) was performed on the NovaSeq 6000 platform targeting 50 million reads per sample. Reads were trimmed to remove adapter sequences using Cutadapt (v2.10) and aligned to the Gencode GRCm38.p6 genome using STAR (v2.7.8a). Gencode vM24 gene annotations were provided to STAR to improve the accuracy of mapping. Quality control on both raw reads and adaptor-trimmed reads was performed using FastQC (v0.11.9) (www.bioinformatics.babraham.ac.uk/projects/fastqc). featureCounts (v2.0.2)15 was used to count the number of mapped reads to each gene.

Differential gene expression analysis
Raw count generated by FeatureCount was imported to R. Genes that were expressed in less than 50% of the samples were excluded from the analysis. The filtered gene count matrix was next used to generate DESeq2 objects with corresponding metadata. Differentially expressed genes between clinical subtypes, molecular subtypes, and immunophenotypes were directly compared using DESeq2 function with design of "~clinical subtype," "~molecular subtype," and "~Immmunophenotype+clinical subtype," respectively. The results from differential gene expression analysis were next fed into GSEA using fgsea based on Hallmark 50 gene pathways or immune cell signatures derived from PangLaoDB database to capture enrichment of immune-related pathways or immune cell subtypes.

Longitudinal comparisons were also performed using DESeq2 when monitoring differential treatment effect within each clinical subtype with or without further separation of the cohort by response. "~Time+patientID" design function was used for all the longitudinal comparisons, thus adjusting for patient specific variants. The results from differential gene expression analysis were then used in GSEA to identify alterations in immune-related pathways and immune cell subtypes.

TCR clonality

After CDR3 region sequencing was extracted using MIXCR, for each sample, TCR clonality was calculated using Shannon diversity index [vegan, R 2.7-2 (*40*)] based on the abundance of different CDR3 regions from TCRβ sequence. Samples with Shannon diversity index of 0 or less than 10 clones detected were removed from future analysis because of low sample quality.

Tumor correlation analysis

Eight continuous immune-related scores were previously calculated on the basis of a patient's tumoral RNA profile (*17*). All eight immune-related scores, except mast cell score, were positively associated with therapy outcomes. However, the immune associated markers in different receptor subtypes seem to have different predictive biology: high dendritic, chemokine, and STAT1 cells/signals best predicted response for TNBC, whereas high B cell combined with low mast cell best predicted pCR in HR^+ $HER2^-$. We performed Pearson correlation analysis between the eight continuous tumoral immune scores with a peripheral blood immune biomarker derived from Hallmark 50 gene pathway or PangLaoDB single-cell immune cell reference. Additionally, a binary immune response variable, $Immune^{\pm}$, was defined by unsupervised clustering of a patient's tumoral immune score (*17*). To identify peripheral immunological differences between $Immune^{\pm}$ patients, differential gene expression analysis was performed using the aforementioned method.

Besides immune associated scores, BC lineage and HR expression were also previously described in these tumors (*17*). A basal/luminal lineage score (derived from BluePrint, Agendia) and ER/PR (estrogen receptor and progesterone receptor) expression calculated by the average of *ESR1* and *PGR* expression in the tumor were also available. We adopted those scores and further tested their association with peripheral immune cell composition with Pearson correlation.

CIBERSORTx immune cell deconvolution

Reference gene matrix for the major PBMC immune cell subtypes was generated by single-cell RNA-sequencing data collected from baseline PBMCs from patients with HR^+ BC (*22*). Eleven immune cell subtypes each containing 500 cells were provided as reference to CIBERSORTx using program default parameter. Immune cell types included four CD8 T cell subtypes [naïve/central memory (CM), early activation, $GZMB^+$ cytotoxic, and $GZMK^+$ cytotoxic], CD4 T cell, B cell, nonclassical and classical monocytes, NK cell, and conventional and plasmacytoid DCs. Bulk RNA-seq data were then deconvoluted using this self-build matrix to derive immune cell composition in absolute mode.

PICS-7 clustering

Raw gene expression matrix was transformed with VST function in DESeq2. The enrichment score of each immune cell gene set from PangLaoDB database was calculated using gene set variance analysis (GSVA) based on the vst-transformed gene matrix [R package (*41*)]. Hierarchical clustering was performed across all the samples collected at baseline and early treatment based on their respective Euclidean distance calculated from the immune cell subtype gene set score. The binary trees were cut by request to generate seven clusters on the basis of their representation on heatmap. Each cluster was annotated by the immune cell supertype (NK cell, T cell, myeloid cell, and B cell) enriched in that cluster.

Prediction model

Logistic regression was performed using the general linear model (GLM) function in R on Chemo+Pembro- or Chemo-treated patients with the formula Response~Clinical subtype + EarlyTreatment 19-gene T cell score|TNBC + Baseline 19-gene T cell score|HR^+ BC + Baseline TCR Shannon + EarlyTreatment-Baseline Leukocyte chemotaxis score|HR^+ BC. Initial training and testing were performed on patients in the pembrolizumab arm. Data from the dostarlimab arm were used for further validation.

Statistical analysis

Individual-level data for experiments where $n < 20$ are presented in data file S1. For testing the differences between groups within a specific time point, two-sided *t* tests or Wilcoxon tests were applied. For cross–time point comparisons, paired *t* tests or Wilcoxon tests were applied. GLM with binomial family was applied to calculate the association between immune cell composition and BC subtype. All correlation analysis was done using Pearson correlation unless otherwise specified. Statistical analysis was performed in R (4.1.3). For individual comparisons, α was set at 0.05. For transcriptome-wide comparisons or multisignature GSEAs, an adjusted (Benjamini-Hochberg) FDR of <0.1 was used. For parametric analyses, data were confirmed to be normally distributed with Shapiro-Wilk test and $P > 0.05$, or a nonparametric equivalent was used if the sample size did not exceed $n = 30$. Survival data were confirmed to be consistent with proportional hazards assumptions in a Cox regression model.

Supplementary Materials

The PDF file includes:
Figs. S1 to S16
Legends for tables S1 to S3
Legend for data file S1

Other Supplementary Material for this manuscript includes the following:
Tables S1 to S3
Data file S1
MDAR Reproducibility Checklist

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Peripheral blood transcriptional profiling predicts tumor subtype and neoadjuvant chemoimmunotherapy outcomes in human breast cancer

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Editor's summary
Although many attempts have been made to predict responses to immunotherapy, they typically rely on tumor tissue specimens, which require invasive collection, or cell-free DNA analysis, which can be costly. Here, Sun *et al.* opted to use peripheral blood and bulk RNA sequencing to define whether a patient with breast cancer was likely to respond to immunotherapy and chemotherapy. The authors identified and validated an immune score that was present before the first treatment or early after treatment initiation and was associated with patient outcomes. The score was associated with markers of cytotoxic CD8 T cell responses, suggesting that these cells might be the effectors driving tumor reduction. Together, these data highlight the value of blood-based biomarkers for cancer and suggest that similar approaches could be applied to other tumor types. —Courtney Malo

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